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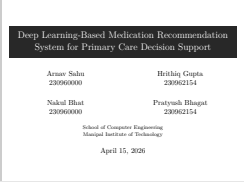
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System Overview

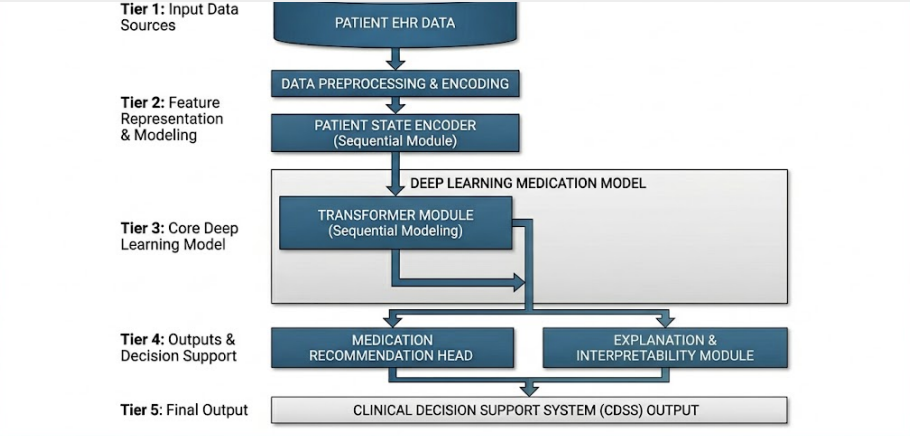
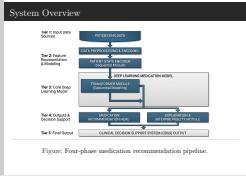


Figure: Four-phase medication recommendation pipeline.



Walk through the figure – four phases: preprocessing, representation learning, recommendation + safety, interpretability.

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Algorithm

Algorithm 1 DL Medication Recommendation

Require: S (symptom text), D (structured data), G (DDI graph)
Ensure: M_{safe}, explanation report E

1: Normalize S: spell correction + NER $\rightarrow$ ontology mapping
2: $E_s \leftarrow \text{ClinicalDistilBERT}(S)$; $\quad E_d \leftarrow \text{MLP}(D)$
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Condensed from the full algorithm – hit the key steps. Good slide to have open during Q&A.

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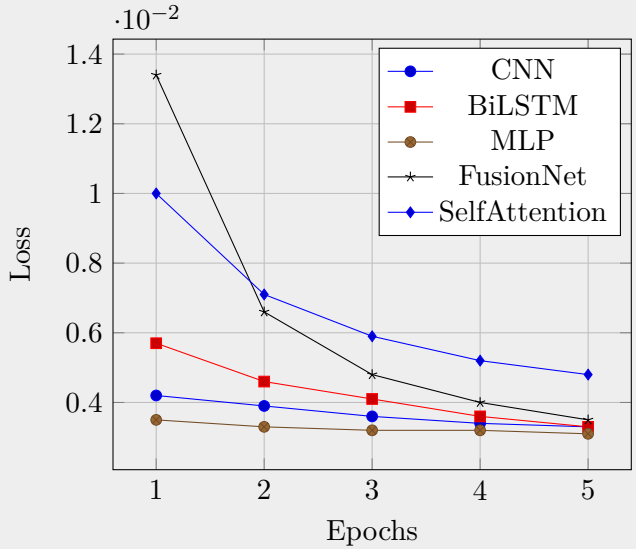
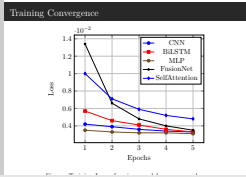
Training Convergence





MLP lowest absolute loss but simpler model. FusionNet starts high, converges fast once text encoder stabilizes – key sign the dual-branch design works.

Multi-Label Classification Performance

Table: Results of the evaluated models across various metrics.

Precision_micro	Recall_micro	F1_micro	Precision_macro
0.8853	0.7561	0.8156	0.0781
0.9031	0.8342	0.8673	0.0858
0.9218	0.8348	0.8762	0.0899
0.9022	0.8382	0.8691	0.0864
0.9252	0.8404	0.8808	0.0898

Subset_Accuracy	ROC_AUC	PRAUC	Precision@K
0.4210	0.9937	0.8372	0.6855
0.5603	0.9968	0.8743	0.7101
0.6184	0.9957	0.8743	0.7076
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FusionNet wins on Jaccard, F1, subset accuracy. MLP competitive on precision but poor subset accuracy – flat features can't capture drug co-occurrence structure. Macro-F1 low across the board due to class imbalance – expected, not a concern.

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Keep this tight. The three adjectives – lightweight, safe, interpretable – are the pitch.

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Future Work

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Good slide to end on – shows the work is a starting point, not a dead end. The LM swap is a quick win worth highlighting.

References I

[1] E. Choi et al. “Doctor AI: Predicting Clinical Events via Recurrent Neural Networks”. In: *MLHC*. 2016.

[2] E. Choi et al. “RETAIN: An interpretable predictive model for healthcare using reverse time attention mechanism”. In: *NeurIPS*. 2016.

[3] W. Lee et al. “SafeDrug: Dual molecular graph encoders for safe drug recommendations”. In: *The Web Conference (WWW)*. 2021.

[4] J. Shang et al. “GAMENet: Graph Augmented Memory Networks for Recommending Medication Combination”. In: *KDD*. 2019.

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